

7 Red blood cells are produced from pluripotent stem cells found in bone marrow.

(a) Which statement about these stem cells is correct?

(1)

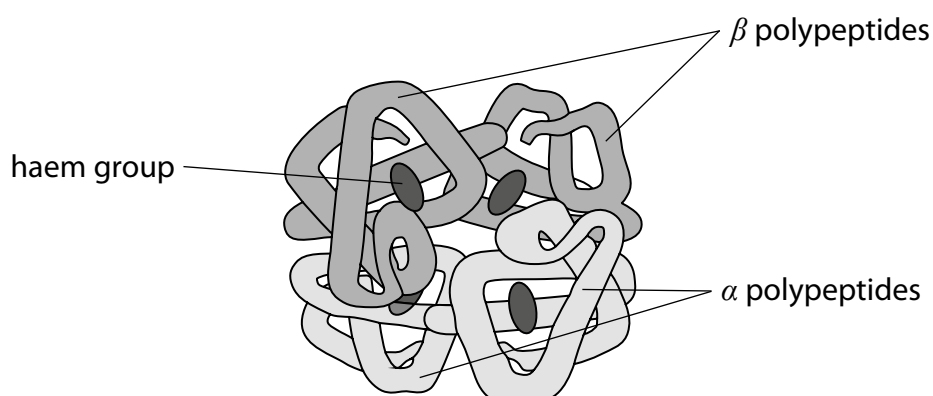
- ☐ A they can produce all types of cell
- ☐ B they can produce all types of cell except extraembryonic cells
- ☐ C they can produce some types of cell
- ☐ D they can produce red blood cells only

(b) Red blood cells contain haemoglobin.

A molecule of haemoglobin is made of four polypeptides. Each polypeptide has a haem group attached to it. The haem group is **not** made of amino acids.

In most adult haemoglobin, there are two α polypeptides and two β polypeptides.

The diagram shows the structure of adult haemoglobin.



Describe the role of the rough endoplasmic reticulum in the synthesis of haemoglobin.

(3)

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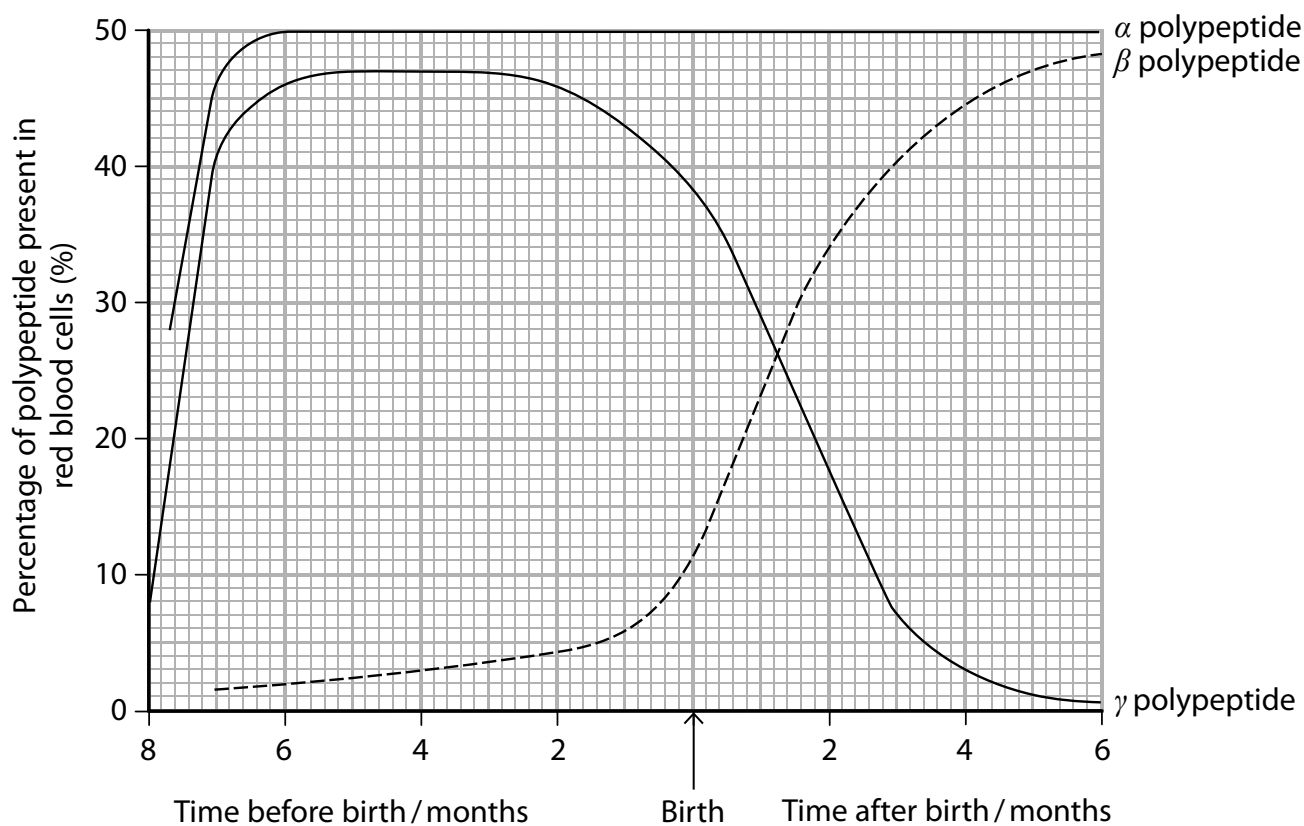
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- (c) Fetal haemoglobin has a similar structure to adult haemoglobin. Fetal haemoglobin has two α polypeptides and two γ polypeptides.

The graph shows the percentage of each polypeptide present in red blood cells in an individual before and after birth.



- (i) Describe the changes in the percentages of polypeptides present in red blood cells. Use the information in the graph to support your answer.

(3)

(ii) Explain how epigenetic modification could result in these changes.

(4)

(Total for Question 7 = 11 marks)